

SALES PERFORMANCE, PROFITABILITY AND BUSINESS INTELLIGENCE DASHBOARD USING POWER BI

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Abstract

This project presents a Power BI-based business intelligence dashboard for retail sales analysis. The solution combines sales, product, customer, salesperson, location and budgeting information into an interactive analytical report. The dashboard includes executive KPIs, sales trend analysis, product performance, customer analysis, salesperson performance, geographic analysis, and budget-versus-actual analysis. Interactive slicers, drill-down and drill-through navigation support detailed exploration of the supplied business data.

I. INTRODUCTION

Retail sales data contains information across transactions, products, customers, locations and salespeople. The Power BI solution provides a centralized view of this information so that business users can examine sales performance, profitability, contribution, geographic variation and budget performance through interactive visuals.

II. PROJECT OBJECTIVES

- Measure sales, orders, quantity, average order value, cost, profit and profit margin.
- Analyze daily, monthly, quarterly and yearly sales trends.
- Identify high- and low-performing products and customers.
- Compare salesperson performance using sales, orders, quantity, profit and AOV.
- Analyze sales by state and location and compare location attributes.
- Compare actual sales with budget using variance and achievement measures.
- Provide interactive filtering, drill-down and drill-through analysis.

III. DATASET AND DATA PREPARATION

The assessment uses the supplied Power BI training dataset containing yearly sales data together with Products, Locations, Customers, Sales People, Budgeting and date-related information. The yearly sales data was consolidated into a common Sales table, while dimension tables were used for descriptive analysis.

IV. DATA MODEL

The report uses a star-schema-oriented model with Sales as the central fact table and dimension tables for Date, Products, Customers, Locations and Sales People. Relationships allow the dimensions to filter sales consistently across dashboard pages.

V. DATE DIMENSION AND DAX MEASURES

A dedicated Date table was created for 2021–2023. It contains Date, Year, Quarter, Month Number, Month, Week, Day and Year-Month. Month is sorted using Month Number for chronological reporting. The model uses reusable DAX measures for sales, profitability, time intelligence and budget analysis.

Category	Measures used in the dashboard
Sales	Total Sales, Total Orders, Total Quantity, Average Order Value, Average Selling Price
Profitability	Total Cost, Total Profit, Profit Margin %, Revenue per Unit, Profit per Unit
Time	Current Year Sales, Previous Year Sales, YoY Sales Growth, YoY Growth %, MTD Sales, YTD Sales, Previous Y
Budget	Total Budget, Budget Variance, Budget Variance %, Budget Achievement %

VI. DASHBOARD DESIGN AND IMPLEMENTATION

The dashboard is organized into seven main analytical pages. The screenshots below document the completed assessment pages.



Fig. 1. Executive Overview showing KPI cards, yearly/monthly sales, product and customer rankings, salesperson performance, state sales and budget comparison.

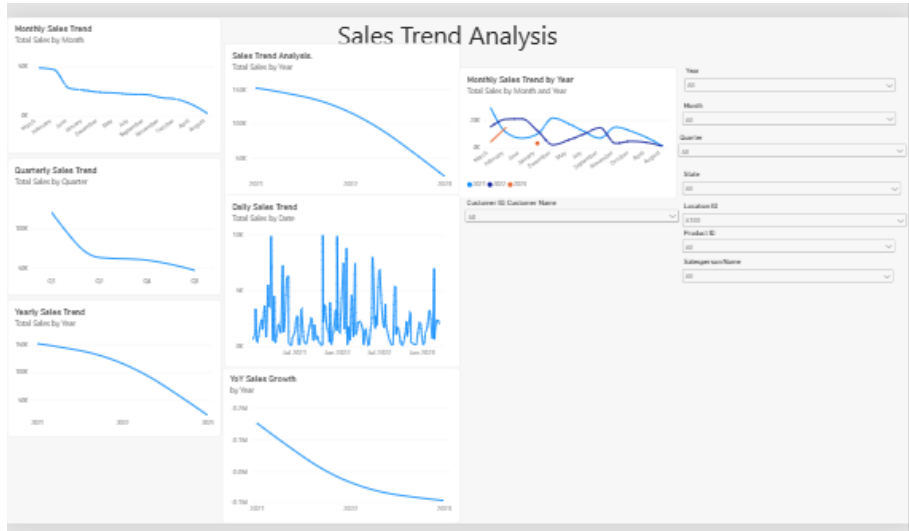


Fig. 2. Sales Trend Analysis page with monthly, yearly, quarterly, daily and YoY trend visuals and interactive slicers.

VII. PRODUCT AND CUSTOMER ANALYSIS

A. Product Performance

The Product Performance page provides top and bottom product views, a product profitability table, sales-versus-profit scatter analysis, product contribution and product-wise sales trends. The page supports filtering by year, product, state and location.

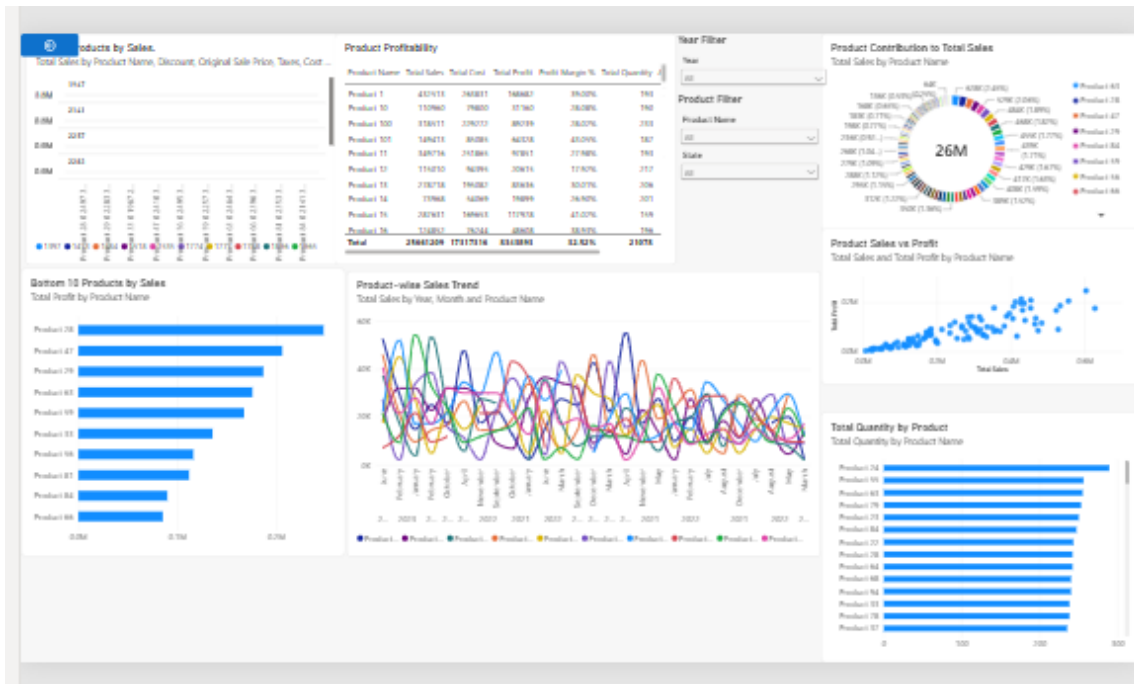


Fig. 3. Product Performance page showing top/bottom products, profitability, contribution, scatter analysis and product sales trend.

B. Customer Analysis

The Customer Analysis page compares customers using sales, contribution, sales and profit, orders and quantity, average order value and sales trends. Top-customer views help identify major contributors and differences in customer performance.

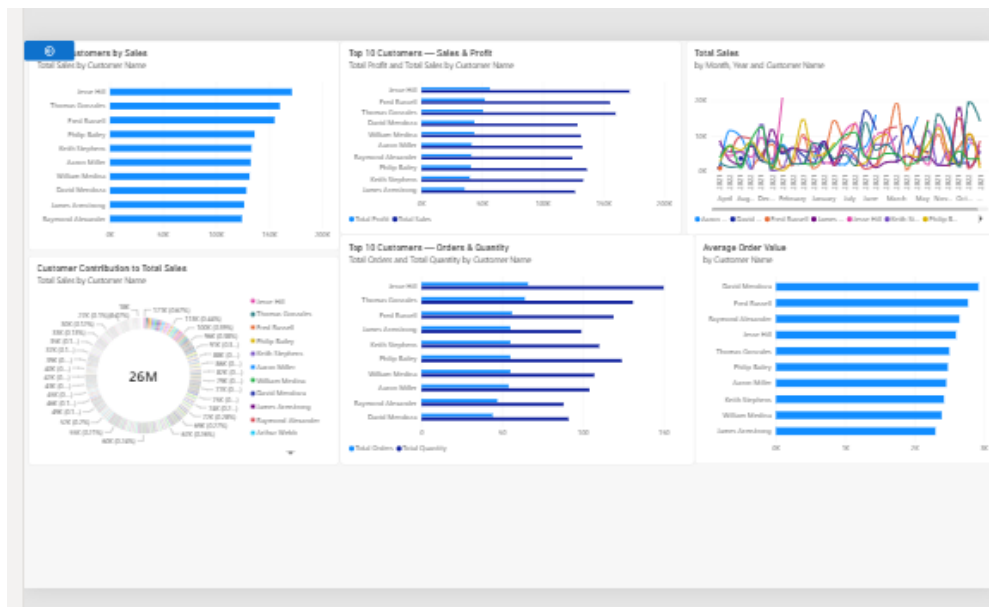


Fig. 4. Customer Analysis page showing customer sales, contribution, sales/profit, orders/quantity, AOV and trend.

VIII. SALESPERSON, GEOGRAPHIC AND BUDGET ANALYSIS

A. Salesperson Performance

Salesperson Performance compares sales, orders, quantity, profit, average order value and contribution. A performance summary and sales trend provide both detailed and time-based analysis.

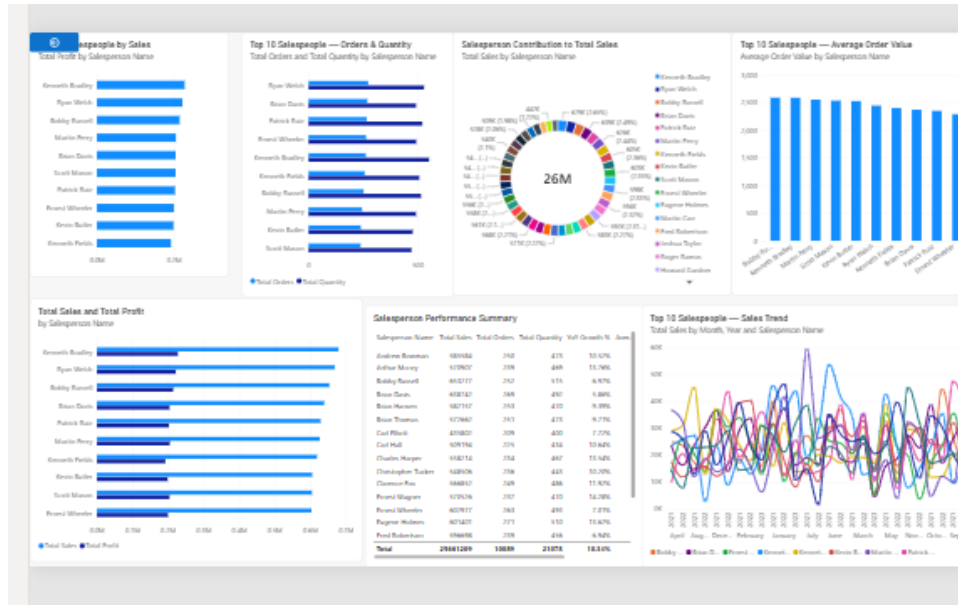


Fig. 5. Salesperson Performance page with rankings, sales/profit, contribution, AOV, summary and trend.

B. Geographic Analysis

Geographic Analysis includes sales by state, sales by location, sales versus population, sales versus households, state ranking, a geographic map and a geographic performance summary matrix.



Fig. 6. Geographic Analysis page showing state/location sales, population and household comparisons, ranking and map area.

1. Power BI Business Intelligence Project: Sales Performance, Profitability & Business Intelligence Dashboard, Assignment Brief.
2. Source retail sales dataset/workbook supplied with the assessment.
3. Microsoft Power BI and DAX documentation used for technical implementation.